



**GIRRAWEEN HIGH SCHOOL**

**TASK 1 2022 (December 2021)**

# **MATHEMATICS**

## **EXTENSION 2**

### **Complex Numbers**

*Time allowed – 90 Minutes*

#### **DIRECTIONS TO CANDIDATES**

- Attempt ALL questions.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Each question attempted is to be marked clearly Question 6, Question 7 etc. Each question is to be returned on a separate page in your answer booklet.
- You may ask for spare answer booklets if you need them.
- For Multiple choice: Fill in the circle corresponding to the correct answer on the multiple choice answer sheet in your answer booklet.
- For Question 9(a) please draw your diagrams on the page of number planes provided in your answer booklet.
- The notation *cis*  $\theta$  is acceptable in this examination.

**Multiple Choice begins on the following page**

**Question 1**

$$(-i)^{2022} =$$

- (A)  $i$                       (B)  $-i$                       (C)  $1$                       (D)  $-1$

**Question 2**

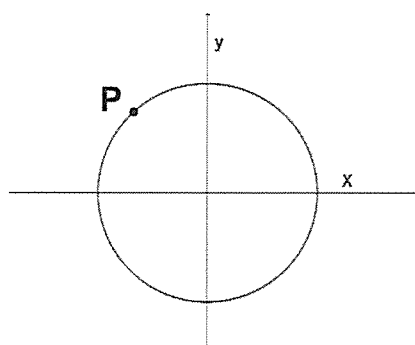
$$4e^{-\frac{\pi}{3}} =$$

- (A)  $2\sqrt{3} - 2i$               (B)  $2 - 2i\sqrt{3}$               (C)  $-2\sqrt{3} + 2i$               (D)  $-2 + 2i\sqrt{3}$

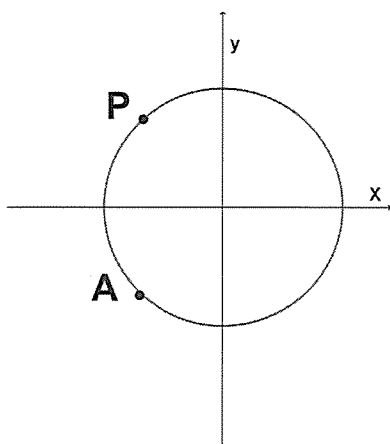
**Question 3**

If  $\overrightarrow{OP} = z$ ,  $\frac{\pi}{2} < \text{Arg } z < \pi$  and  $|z| > 1$  (see diagram),

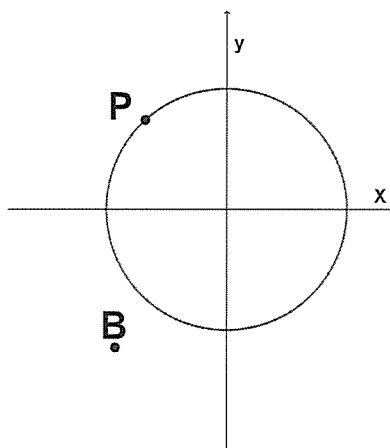
then  $\sqrt{z}$  would be



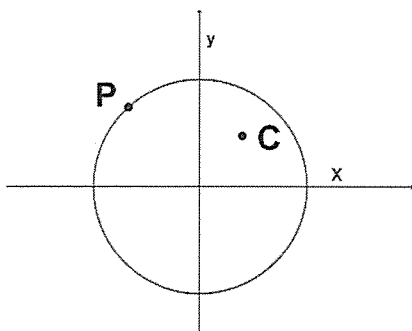
(A)  $\overrightarrow{OA}$



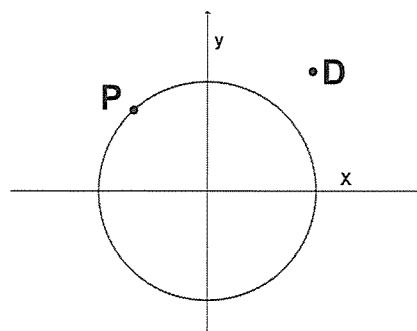
(B)  $\overrightarrow{OB}$



(C)  $\overrightarrow{OC}$



(D)  $\overrightarrow{OD}$



*Multiple Choice continues on the following page*

**Multiple Choice (continued)**

**Question 4**

The solutions to  $z^2 - 2z + 5 = 0$  are

- (A)  $z = 1 \pm 2i$       (B)  $z = -1 \pm 2i$       (C)  $z = 1 \pm 4i$       (D)  $z = 1 \pm 4i$

**Question 5**

$$z \times \frac{\bar{z}}{|z|^2} =$$

- (A)  $\frac{z}{\bar{z}}$       (B)  $\frac{1}{z}$       (C)  $z$       (D)  $1$

**Question 6 (20 Marks – show all of your working in your answer booklet)** **Marks**

(a) If  $z = -\sqrt{3} - i$  and  $w = 1 - i$

(i) Find  $\frac{z}{w}$  in Cartesian form. **2**

(ii) Convert  $z$  and  $w$  to modulus/argument form. **4**

(iii) Find  $\frac{z}{w}$  in modulus/argument form. **2**

(iv) Use your result to find the exact value of  $\cos\left(-\frac{7\pi}{12}\right)$ . **2**

(b) (i) By using  $(x + iy)^2 = 15 - 8i$ , find the square roots of  $15 - 8i$  in Cartesian form. **3**

(ii) Hence solve  $z^2 + (2 - i)z + (i - 3) = 0$ . **3**

(c) Find the five fifth roots of  $-16\sqrt{3} + 16i$ . Leave your answers in modulus/argument form. **4**

**Question 7 (7 Marks – show all of your working in your answer booklet)**

(a) If the roots of the polynomial equation  $4z^3 - 13z^2 + 40z + 39 = 0$  are  $\alpha, \beta$  and  $\lambda$ , find  $\alpha^2 + \beta^2 + \lambda^2$  WITHOUT finding  $\alpha, \beta$  or  $\lambda$ . **2**

(b) Using your result from (a), state why  $4z^3 - 13z^2 + 40z + 39 = 0$  has exactly one real root. **2**

(c) If one of the roots of  $4z^3 - 13z^2 + 40z + 39 = 0$  is  $z = 2 + 3i$ , solve  $4z^3 - 13z^2 + 40z + 39 = 0$  over the complex field. **3**

*Examination continues on the following page*

**Question 8 (12 Marks – show all of your working in your answer booklet)** **Marks**

(a) Sketch the following curves and regions on separate complex planes.

(i)  $|z - 3 + i| = 2$  **2**

(ii)  $\text{Arg}(z + 1) = -\frac{2\pi}{3}$  **2**

(iii)  $|z| = 4$  and  $-\frac{\pi}{4} < \text{Arg} z \leq \frac{\pi}{4}$  **3**

(b) The locus  $\text{Arg}\left(\frac{z-3-i}{z+1-i}\right) = \frac{\pi}{4}$  represents part of a circle. Find the centre and radius of this circle and write the Cartesian equation for the locus. **5**

**Question 9 (10 Marks – show all of your working in your answer booklet)**

(a) If  $\overrightarrow{OA} = z$  and  $\overrightarrow{OB} = w$  with  $0 \leq \text{Arg} z \leq \frac{\pi}{2}$ ,

$0 \leq \text{Arg} w \leq \frac{\pi}{2}$  and both  $|z|$  and  $|w| > 1$  (see diagram),

sketch on the Argand diagrams provided for 9(a) in your answer booklet

(Note that these vectors all start from O)

(i) C so that  $\overrightarrow{OC} = z + w$  **1**

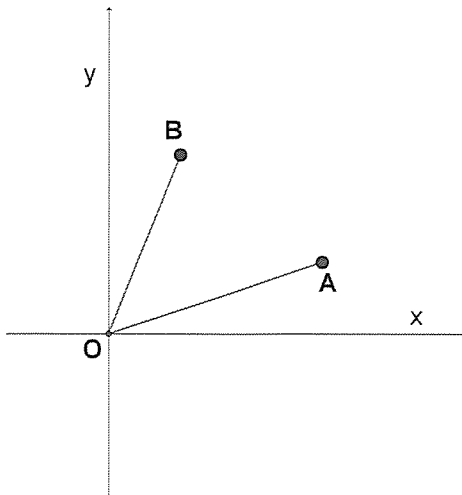
(ii) D so that  $\overrightarrow{OD} = w - z$  **1**

(iii) E so that  $\overrightarrow{OE} = z + 1$  **1**

(iv) F so that  $\overrightarrow{OF} = z - 2i$  **1**

(v) G so that  $\overrightarrow{OG} = -iz$  **1**

(vi) H so that  $\overrightarrow{OH} = wcis\frac{\pi}{4}$  **1**

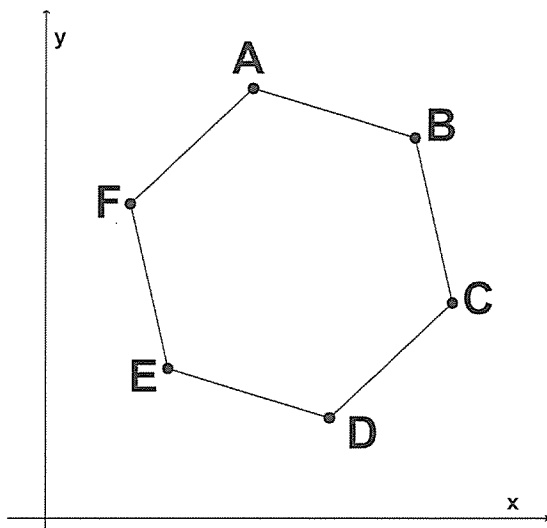


*Question 9 continues on the following page*

**Question 9 (continued)**

**Marks**

(b)  $ABCDEF$  is a regular hexagon in Quadrant 1 of the complex plane  
(see diagram)



If  $\overrightarrow{OA} = z_1$ ,  $\overrightarrow{OB} = z_2$ ,  $\overrightarrow{OC} = z_3$ ,  $\overrightarrow{OD} = z_4$ ,  $\overrightarrow{OE} = z_5$  and  $\overrightarrow{OF} = z_6$

- (i) Explain why  $\text{Arg} \left( \frac{z_2 - z_3}{z_2 - z_5} \right) = \frac{\pi}{3}$  2
- (ii) Show that  $\left( \frac{z_2 - z_5}{z_3 - z_1} \right)$  is entirely imaginary. 2

**Question 10 (13 Marks – show all of your working in your answer booklet)**

- (a)(i) Solve  $\cos 4\theta = 0, 0 \leq \theta \leq 2\pi$ . 2
- (ii) Using DeMoivre's Theorem and the expansion of  $(\cos\theta + i\sin\theta)^4$  3  
show that  $\cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1$ .
- (iii) Hence find the exact value of  $\cos \frac{\pi}{8}$ . 2  
(you may leave your answer as a fourth root).
- (b)(i) Using Euler's Theorem, show that  $e^{ni\theta} + e^{-ni\theta} = 2 \cos n\theta$ . 2
- (ii) Hence show that  $\cos^4\theta = \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8}$ . 2
- (iii) Hence find the exact value of  $\cos \frac{\pi}{8}$  2  
(you may leave your answer as a fourth root).

*Examination continues on the following page*

**Question 11 (7 Marks – show all of your working in your answer booklet)** **Marks**

**(a)** Solve  $z^9 - 1 = 0$ . **2**

**(b)** If  $w$  is the root of  $z^9 - 1 = 0$  with the smallest positive argument, show that

**(i)**  $w + w^2 + w^3 + w^4 + w^5 + w^6 + w^7 + w^8 = -1$  **1**

(you may assume that  $w^2, w^3$  etc. are also roots of  $z^9 - 1 = 0$ ).

**(ii)** Show that  $(w^2 + w^7)(w^4 + w^5)(w + w^8) = -1$  **2**

**(c)** Hence show that  $\cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9} = \frac{1}{8}$ . **2**

(You may assume that if  $z = \text{cis}\theta$ ,  $z^n + \frac{1}{z^n} = 2\cos n\theta$ )

**END OF EXAMINATION!!!**



# GIRRAWEE HIGH SCHOOL

## MATHEMATICS EXTENSION 2

### 2022 HSC Task 1 (December 2021)

Student No: \_\_\_\_\_

*Solution [Final]*

This Booklet contains the answer sheet for Part A and Writing Booklet for Part B.

#### Part A ANSWER SHEET

Select the alternative A, B, C or D that best answers the question.

|    |   |                                  |   |                                  |   |                                  |   |                                  |
|----|---|----------------------------------|---|----------------------------------|---|----------------------------------|---|----------------------------------|
| 1. | A | <input type="radio"/>            | B | <input type="radio"/>            | C | <input type="radio"/>            | D | <input checked="" type="radio"/> |
| 2. | A | <input type="radio"/>            | B | <input checked="" type="radio"/> | C | <input type="radio"/>            | D | <input type="radio"/>            |
| 3. | A | <input type="radio"/>            | B | <input type="radio"/>            | C | <input checked="" type="radio"/> | D | <input type="radio"/>            |
| 4. | A | <input checked="" type="radio"/> | B | <input type="radio"/>            | C | <input type="radio"/>            | D | <input type="radio"/>            |
| 5. | A | <input type="radio"/>            | B | <input type="radio"/>            | C | <input type="radio"/>            | D | <input checked="" type="radio"/> |

#### Instructions

- If you need more paper for Part B, please ask your supervisor.
- Write your name on every booklet you use.
- Write on both sides of each sheet of paper.

Total number of booklets used \_\_\_\_\_.

# Solution: Complex Numbers Test: Dec. 2021 p.1

## Multiple Choice:

(1)  $(-i)^{2022}$

$$= (-i)^{2020} \times (-i)^2$$

$$= 1 \times -1$$

$$= -1. \quad \textcircled{D}$$

(2)  $4e^{-\frac{\pi}{3}}$

$$= 4 \left[ \cos\left(-\frac{\pi}{3}\right) + i \sin\left(-\frac{\pi}{3}\right) \right]$$

$$= 4 \left[ \frac{1}{2} - i \frac{\sqrt{3}}{2} \right]$$

$$= 2 - 2i\sqrt{3}. \quad \textcircled{B}$$

(3)  $|\sqrt{z}| < |z|$  if  $|z| > 1$

so  $\sqrt{z}$  is INSIDE circle.

If  $\frac{\pi}{2} < \text{Arg } z < \pi$  then

$$0 < \text{Arg}(\sqrt{z}) < \frac{\pi}{2}$$

as  $\text{Arg}(\sqrt{z}) = \frac{1}{2} \text{Arg } z. \quad \textcircled{C}$

(4)  $z^2 - 2z + 5 = 0$

$$(z-1)^2 + 4 = 0$$

$$[z-1-2i][z-1+2i] = 0$$

$$z = 1 \pm 2i. \quad \textcircled{A}$$

[Could also be done using the quadratic formula].

(5)  $z \times \frac{\bar{z}}{|z|^2}$

$$= \frac{z\bar{z}}{|z|^2}$$

$$= \frac{|z|^2}{|z|^2}$$

$$= 1 \quad \textcircled{D}$$

Q. (6)(a)(i)  $\frac{z}{w}$

$$= \frac{\sqrt{3}-i}{1-i} \times \frac{(1+i)}{(1+i)}$$

$$= \frac{-\sqrt{3}-i\sqrt{3}-i+1}{2}$$

$$= \frac{(1-\sqrt{3})-i(\sqrt{3}+1)}{2}$$

$$= \left( \frac{1-\sqrt{3}}{2} \right) + i \left( \frac{-\sqrt{3}-1}{2} \right)$$

(ii)  $z = 2 \text{cis}\left(-\frac{5\pi}{6}\right)$  4

$w = \sqrt{2} \text{cis}\left(-\frac{\pi}{4}\right)$

(iii)  $\frac{z}{w} = \sqrt{2} \text{cis}\left(-\frac{7\pi}{12}\right)$  2

(b)(i)  $(x+iy)^2 = 15 - 8i, x, y \text{ REAL}$

$$(x^2 - y^2) + 2ixy = 15 - 8i$$

Equating reals, Equating imaginaries

$$x^2 - y^2 = 15 \quad (1) \quad 2xy = -8$$

$$y = -\frac{4}{x} \quad (2)$$

Sub. (2) in (1):

$$x^2 - \left(-\frac{4}{x}\right)^2 = 15$$

$$x^2 - \frac{16}{x^2} = 15$$

$$x^4 - 15x^2 - 16 = 0$$

$$(x^2 - 16)(x^2 + 1) = 0$$

As  $x$  is real,  $x^2 - 16 = 0$   
 $x = \pm 4$

As  $y = -\frac{4}{x}$  3

$$y = -\frac{4}{4} \text{ or } -\frac{4}{-4}$$

$$= -1 \text{ or } 1.$$

So  $\sqrt{15-8i}$

$$= \pm(4-i).$$

(iv) Equating real parts of (i) & (iii)

$$\frac{1-\sqrt{3}}{2} = \sqrt{2} \cos\left(-\frac{7\pi}{12}\right) \quad \underline{\underline{2}}$$

$$\therefore \cos\left(-\frac{7\pi}{12}\right) = \frac{1-\sqrt{3}}{2\sqrt{2}}$$

Complex Numbers: p.2

$$Q.(6)(b)(i) \quad z^2 + (2-i)z + (i-3) = 0$$

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-2+i \pm \sqrt{15-8i}}{2 \times 1}$$

$$= \frac{-2+i \pm (4-i)}{2} \text{ [from (b)(i)]}$$

Note:  $b^2 - 4ac$

$$= (2-i)^2 - 4 \times 1 \times (i-3)$$

$$= 4 - 4i - 1 - 4i + 12$$

$$= 15 - 8i$$

3

$$z = \frac{-2+i+4-i}{2}$$

$$\text{or } z = \frac{-2+i-4+i}{2}$$

$$z = 1 \quad \text{or } z = -3+i$$

$$(c) \quad -16\sqrt{3} + 16i$$

$$= 32 \operatorname{cis} \frac{5\pi}{6}$$

$\therefore$  The five 5th roots of  $-16\sqrt{3} + 16i$

$$= 2 \operatorname{cis} \left[ \frac{\pi}{6} + \frac{2k\pi}{5}, k=0,1,2,3,4 \right]$$

4

$$\text{or } = 2 \operatorname{cis} \frac{\pi}{6}, 2 \operatorname{cis} \frac{17\pi}{30}, 2 \operatorname{cis} \frac{29\pi}{30}, 2 \operatorname{cis} \frac{41\pi}{30}, 2 \operatorname{cis} \frac{53\pi}{30}$$

$$\text{or } = 2 \operatorname{cis} \frac{\pi}{6}, 2 \operatorname{cis} \frac{17\pi}{30}, 2 \operatorname{cis} \frac{29\pi}{30}, 2 \operatorname{cis} \frac{-19\pi}{30}, 2 \operatorname{cis} \frac{-7\pi}{30}$$

# Complex Numbers Solutions [continued] p.3

$$\begin{aligned} \text{Q. (7) (a)} \quad & \alpha^2 + \beta^2 + \lambda^2 \\ &= (\alpha + \beta + \lambda)^2 - 2(\alpha\beta + \alpha\lambda + \beta\lambda) \\ &= \left(\frac{13}{4}\right)^2 - 2 \times \frac{40}{4} \\ &= \underline{\underline{-\frac{151}{16}}} \end{aligned}$$

(b) As  $\alpha^2 + \beta^2 + \lambda^2 < 0$ ,  $4z^3 - 13z^2 + 40z + 39 = 0$  must have at least 1 non-real root as a real number squared  $\geq 0$ . 2

As  $4z^3 - 13z^2 + 40z + 39 = 0$  has all real co-efficients, the CONJUGATE of any non-real root must ALSO be a solution.

Hence  $4z^3 - 13z^2 + 40z + 39 = 0$  has 1 real & 2 non-real roots.

(c) One root of  $4z^3 - 13z^2 + 40z + 39 = 0$  is  $z = 2 + 3i$ .  
 $\therefore$  Another root is  $\bar{z} = 2 - 3i$ .

$$\text{So by } \alpha + \beta + \lambda = -\frac{b}{a}.$$

$$\begin{aligned} (2+3i) + (2-3i) + \lambda &= \frac{13}{4} \\ \lambda &= \underline{\underline{-\frac{3}{4}}} \end{aligned}$$

Solutions to  $4z^3 - 13z^2 + 40z + 39 = 0$  are  
 $z = 2 \pm 3i, -\frac{3}{4}$

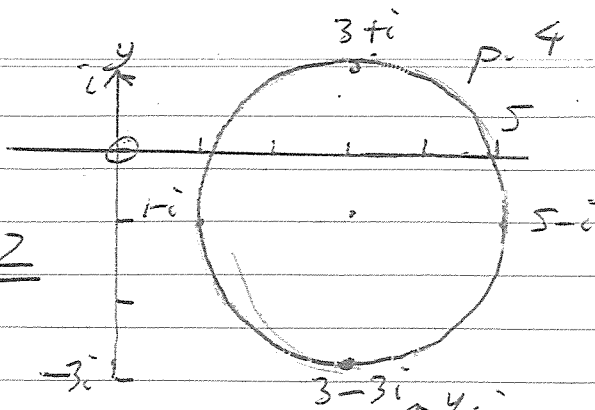
Q. (8) (a)

(i)  $|z - 3 + i| = 2$ .

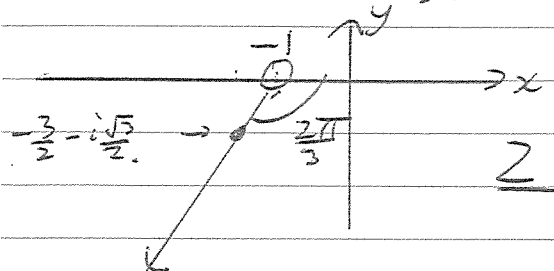
Circle centre  $3 - i$ ,  $r = 2$ .

Equation:

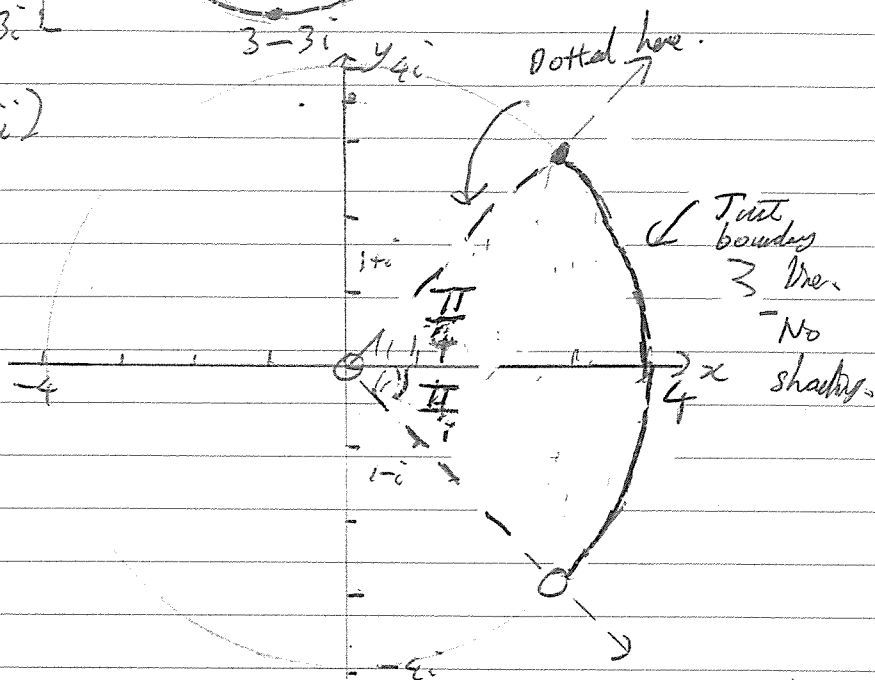
$(x - 3)^2 + (y + 1)^2 = 4$ . 2



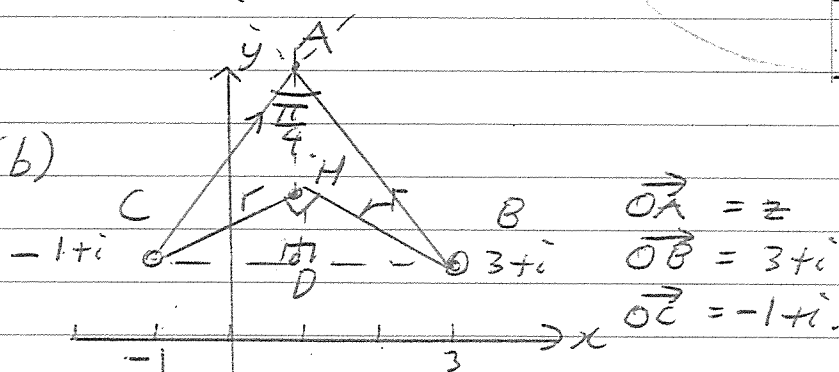
(ii)  $\text{Arg}(z + 1) = -\frac{2\pi}{3}$ .



(iii)



Q. (8) (b)



$\vec{OA} = z$

$\vec{OB} = 3 + i$

$\vec{OC} = -1 + i$

$\vec{BA} = z - 3 - i$   $\frac{\pi}{2} \leq \text{Arg } \vec{BA} < \pi$

$\vec{CA} = z + 1 - i$   $0 \leq \text{Arg } \vec{CA} < \frac{\pi}{2}$

$\therefore \text{Arg} \left( \frac{z - 3 - i}{z + 1 - i} \right) = \angle CAB = \frac{\pi}{4}$

Let circle centre = H.

x co-ordinate of H = 1.

$\angle CHB = \frac{\pi}{2}$  [L at centre =  $2 \times$  L at circumference subtended by arc CB].

$CH = BH = r$  [radius of circle]

By Pythagoras on  $\triangle CHB$ ,  $r^2 + r^2 = (CB)^2 = 4$ .  
 $r = 2\sqrt{2}$ .

Let D be midpoint of

$HCB = 1 + i$

Using  $\triangle CHO$ :

$2\sqrt{2}^2 = (2\sqrt{2})^2 - 2^2 = 4$ .

$CH^2 = 4$ . 5

$\therefore H = 1 + 3i$

Locus is

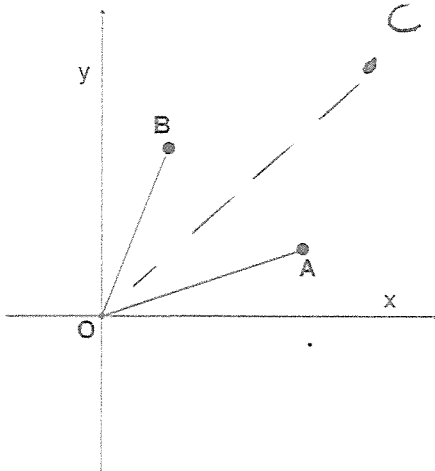
$(x - 1)^2 + (y - 3)^2 = 8$ ,

$y > 1$ .

P.5

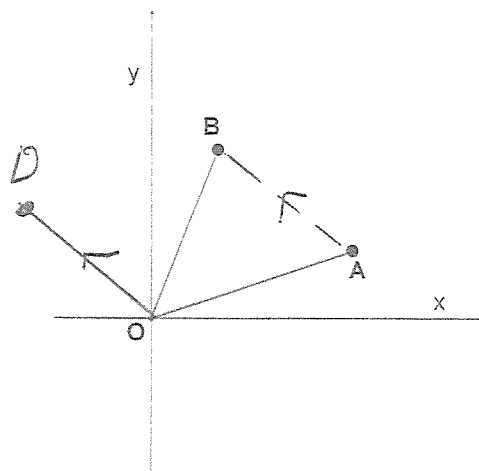
Answer sheet for Question 9(a)

(i)



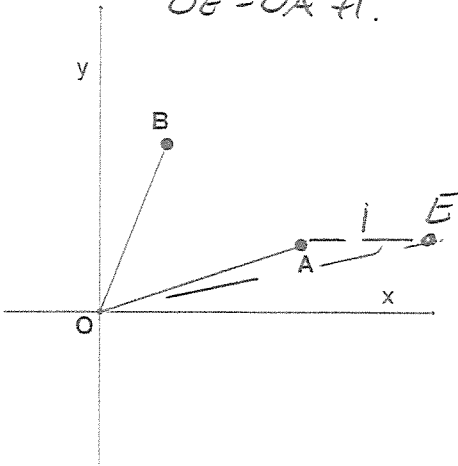
(ii)

$$\vec{OD} = \vec{AB} = w - z.$$

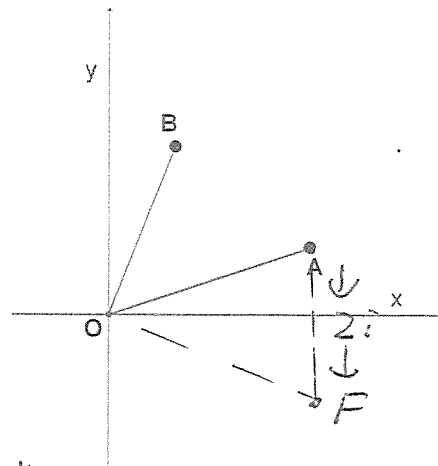


(iii)

$$\vec{OE} = \vec{OA} + 1.$$

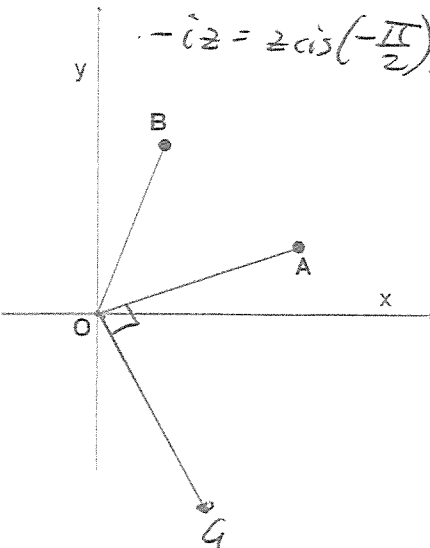


(iv)

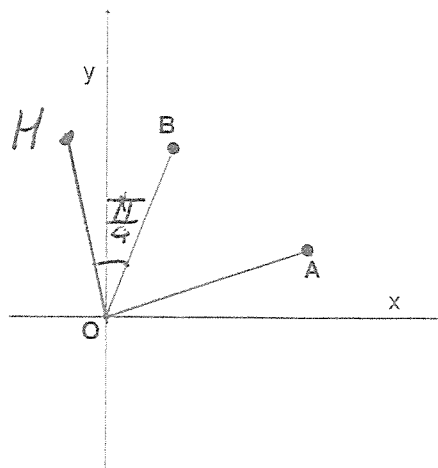


(v)

$$-iz = z \cos\left(-\frac{\pi}{2}\right).$$

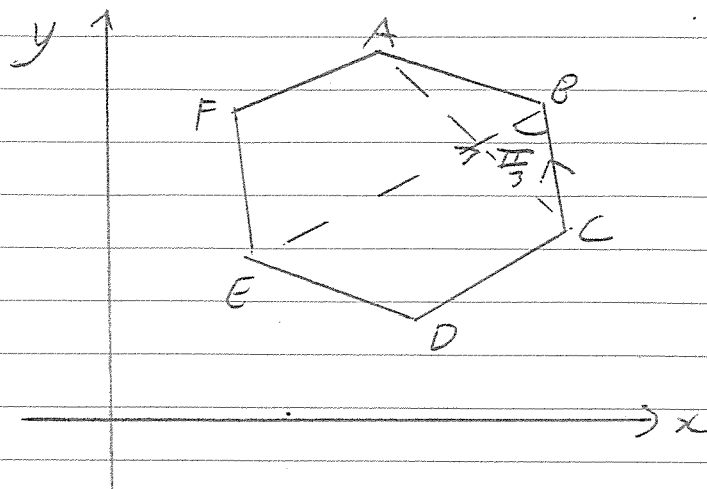


(vi)



Q. (9) (b)

(p. 6)



$$(i) \operatorname{Arg} \left( \frac{z_2 - z_3}{z_2 - z_5} \right) = \angle EBC \text{ as } z_2 - z_3 = \overrightarrow{CB} \text{ \& } z_2 - z_5 = \overrightarrow{EB}.$$

$$\angle ABC = \frac{2\pi}{3} \text{ (interior } \angle \text{ of regular hexagon).} \quad \underline{2}$$

As EB bisects  $\angle ABC$ ,  $\angle EBC = \frac{\pi}{3}$

$$(ii) z_3 - z_1 = \overrightarrow{AC}$$

Letting G be the point of intersection of AC & EB:

Using  $\triangle BCO$

$$\angle BCA = \frac{\pi}{6} \text{ [ } \angle \text{ opposite = sides in isosceles } \triangle ABC \text{]}$$

Using  $\triangle BCG$

$$\angle BGC = \frac{\pi}{2} \text{ [ } \angle \text{ sum of } \triangle BGC \text{]} \quad \underline{2}$$

$$\therefore \text{ As } \operatorname{Arg} \left( \frac{z_2 - z_5}{z_3 - z_1} \right) = \angle BGC.$$

$$\operatorname{Arg} \left( \frac{z_2 - z_5}{z_3 - z_1} \right) = \frac{\pi}{2}.$$

$$\frac{z_2 - z_5}{z_3 - z_1} \text{ is entirely imaginary.}$$

p. 7

Q. (10) (a) (i)  $\cos 4\theta = 0$

$$4\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \frac{9\pi}{2}, \frac{11\pi}{2}, \frac{13\pi}{2}, \frac{15\pi}{2}$$

$$\theta = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}, \frac{9\pi}{8}, \frac{11\pi}{8}, \frac{13\pi}{8}, \frac{15\pi}{8} \quad \underline{2}$$

(ii) By de Moivre

$$(\cos \theta + i \sin \theta)^4$$

$$= \cos 4\theta + i \sin 4\theta$$

$$\cos^4 \theta + 4i \cos^3 \theta \sin \theta - 6 \cos^2 \theta \sin^2 \theta - 4i \cos \theta \sin^3 \theta + \sin^4 \theta = \cos 4\theta + i \sin 4\theta \quad \text{--- (1)}$$

Equating real parts,

$$\cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta$$

$$= \cos^4 \theta - 6 \cos^2 \theta (1 - \cos^2 \theta) + (1 - \cos^2 \theta)^2$$

$$= 8 \cos^4 \theta - 8 \cos^2 \theta + 1, \text{ as required.} \quad \underline{3}$$

(iii) Hence  $\frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}$  etc. are solutions to  $\cos 4\theta = 0$   
i.e.  $8 \cos^4 \theta - 8 \cos^2 \theta + 1 = 0$

$$\cos^2 \theta = \frac{8 \pm \sqrt{(-8)^2 - 4 \times 8 \times 1}}{2 \times 8}$$

$$\cos^2 \theta = \frac{2 \pm \sqrt{2}}{4}$$

$$\cos^2 \theta = \pm \sqrt{\frac{2 \pm \sqrt{2}}{4}} \quad \underline{2}$$

As  $\frac{\pi}{8}$  is in  $Q_1$ ,  $\cos \frac{\pi}{8}$  will be positive &

will be  $\sqrt{\frac{2+\sqrt{2}}{4}}$  or  $\sqrt{\frac{2-\sqrt{2}}{4}}$ .

$$\text{As } \cos \frac{\pi}{8} > \cos \frac{3\pi}{8}, \cos \frac{\pi}{8} = \sqrt{\frac{2+\sqrt{2}}{4}}$$

Q. 10)(b)(i) By Euler's theorem,

$$e^{ni\theta} + e^{-ni\theta} = (\cos n\theta + i\sin n\theta) + (\cos(-n\theta) + i\sin(-n\theta)) \quad \underline{2}$$

$$= \cos n\theta + i\sin n\theta + \cos n\theta - i\sin n\theta \quad [\text{as } \cos \text{ even, } \sin \text{ odd}]$$

$$= 2\cos n\theta \quad \text{as required.}$$

(ii) Hence  $[2\cos\theta]^4 = (e^{i\theta} + e^{-i\theta})^4$

$$16\cos^4\theta = e^{4i\theta} + 4e^{2i\theta} + 6 + 4e^{-2i\theta} + e^{-4i\theta} \quad \underline{2}$$

$$= (e^{4i\theta} + e^{-4i\theta}) + 4(e^{2i\theta} + e^{-2i\theta}) + 6$$

$$16\cos^4\theta = 2\cos 4\theta + 8\cos 2\theta + 6$$

$$\cos^4\theta = \frac{1}{8}\cos 4\theta + \frac{1}{2}\cos 2\theta + \frac{3}{8} \quad \text{as required.}$$

(iii) Hence  $\cos^4\frac{\pi}{8} = \frac{1}{8}\cos\frac{\pi}{2} + \frac{1}{2}\cos\frac{\pi}{4} + \frac{3}{8}$

$$\cos^4\frac{\pi}{8} = \frac{1}{2\sqrt{2}} + \frac{3}{8} \quad \underline{2}$$

$$\cos^4\frac{\pi}{8} = \frac{2\sqrt{2} + 3}{8}$$

$$\cos\frac{\pi}{8} = \sqrt[4]{\frac{2\sqrt{2} + 3}{8}}$$

as cos positive in  $Q_1$ , just positive  $\sqrt[4]{}$ .

[Note: Can also do  $\cos^4\frac{\pi}{8} = \frac{4\sqrt{2} + 6}{16}$

$$\cos\frac{\pi}{8} = \sqrt[4]{\frac{4\sqrt{2} + 6}{16}}$$

p. 9

Q.11)(a)  $z^9 - 1 = 0$

$z = \text{cis } \frac{2k\pi}{9}, k = 0, 1, 2, 3, 4, 5, 6, 7, 8.$

(b)(i)  $z^9 - 1 = 0$

By  $\alpha + \beta + \gamma + \lambda = -\frac{b}{a}$

$1 + w + w^2 + w^3 + w^4 + w^5 + w^6 + w^7 + w^8 = 0$

$w + w^2 + w^3 + w^4 + w^5 + w^6 + w^7 + w^8 = -1.$

Note: Can also use

$z^9 - 1 = 0$

$(z-1)(1+z+z^2+z^3+z^4+z^5+z^6+z^7+z^8) = 0$

sub in  $w$   
 $(w-1)(1+w+w^2+w^3+w^4+w^5+w^6+w^7+w^8) = 0$

As  $w \neq 1$ ,  $1 + w + w^2 + w^3 + w^4 + w^5 + w^6 + w^7 + w^8 = 0.$

(ii)  $(w^2 + w^7)(w^4 + w^5)(w + w^8)$

$= (w^6 + w^7 + w^{11} + w^{12})(w + w^8)$

$= w^7 + w^{14} + w^8 + w^{15} + w^{12} + w^{19} + w^{13} + w^{20}$

$= w^7 + w^5 + w^8 + w^6 + w^3 + w + w^4 + w^2$

$= -1$  (using f(i)).

(c) As  $w + w^8 = \text{cis } \frac{2\pi}{9} + \text{cis } \frac{16\pi}{9}$

$= 2 \cos \frac{2\pi}{9}$

Similarly  $w^2 + w^7 = 2 \cos \frac{4\pi}{9}$  &  $w^4 + w^5 = 2 \cos \frac{8\pi}{9}$

$= -2 \cos \frac{\pi}{9}$

By  $(w^2 + w^7)(w^4 + w^5)(w + w^8)$

$= -1.$

$2 \cos \frac{4\pi}{9} \times 2 \cos \frac{\pi}{9} \times 2 \cos \frac{2\pi}{9}$

$= -1$

$-8 \cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9}$

$= -1 \therefore -8$

$\cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9}$

$= \frac{1}{8}$

END OF SOLUTION!!!